SCHOOL OF COMPUTING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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10214AD701- MAJOR PROJECT INHOUSE

“SPATIOTEMPORAL TRAFFIC PREDICTION AND RECOMMENDATION USING CTNS”

SEMESTER END REVIEW

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• Introduction

- **Brief intro to the project topic**
- This project focuses on predicting traffic conditions by analyzing both spatial and temporal traffic patterns using CTNS.
- It also provides route recommendations to help users avoid congestion and travel efficiently.It also provides route recommendations to help users avoid congestion and travel efficiently.
- **Motivation behind choosing the topic**
- Increasing traffic congestion in urban areas causes delays, fuel wastage, and environmental pollution. The project was chosen to develop an intelligent system that can predict future traffic and suggest better routes in real time.
- Rapid growth in the number of vehicles and increasing urbanization have made traffic congestion a major issue, motivating the development of a smarter and more efficient traffic prediction system.
- **Problem Statement**
- This project aims to build a spatiotemporal prediction model using CTNS to improve traffic forecasting and route recommendation.
- A reliable intelligent traffic prediction system is required to analyze real-time and historical traffic data, reduce congestion, and provide users with efficient route recommendations.

• Objective of the Project

- **Main objectives**
- To develop a CTNS-based model for accurate spatiotemporal traffic prediction by analyzing historical and real-time traffic data.
- To identify congestion-prone areas and forecast future traffic conditions in advance.
- To provide intelligent route recommendations that minimize travel time, fuel consumption, and traffic delays.
- To improve the efficiency of urban traffic management systems and support safer, faster transportation.

- **What the project aims to solve or improve**
- The project aims to solve the problem of inaccurate traffic prediction and inefficient route planning in urban areas.
- It improves traffic management by forecasting congestion in advance and recommending optimal routes, thereby reducing travel time, fuel consumption, and traffic-related delays.

• Literature Review / Existing System

- **Existing systems/methods are out there**
- Statistical models such as ARIMA are used to predict traffic flow based on past data.
- Deep learning models like CNN and LSTM are applied for traffic forecasting.
- GPS-based navigation systems provide real-time route suggestions.
- Applications such as Google Maps and Waze are widely used for traffic monitoring and navigation.

- **Their limitations or problems**
- Existing systems may not accurately capture both spatial and temporal traffic patterns.
- They often fail to predict sudden congestion, accidents, or peak-hour traffic.
- Route recommendations are usually based only on current traffic conditions, not future traffic.
- Prediction accuracy decreases in complex urban road networks, leading to delays and inefficient routing.

• Literature Review Summary

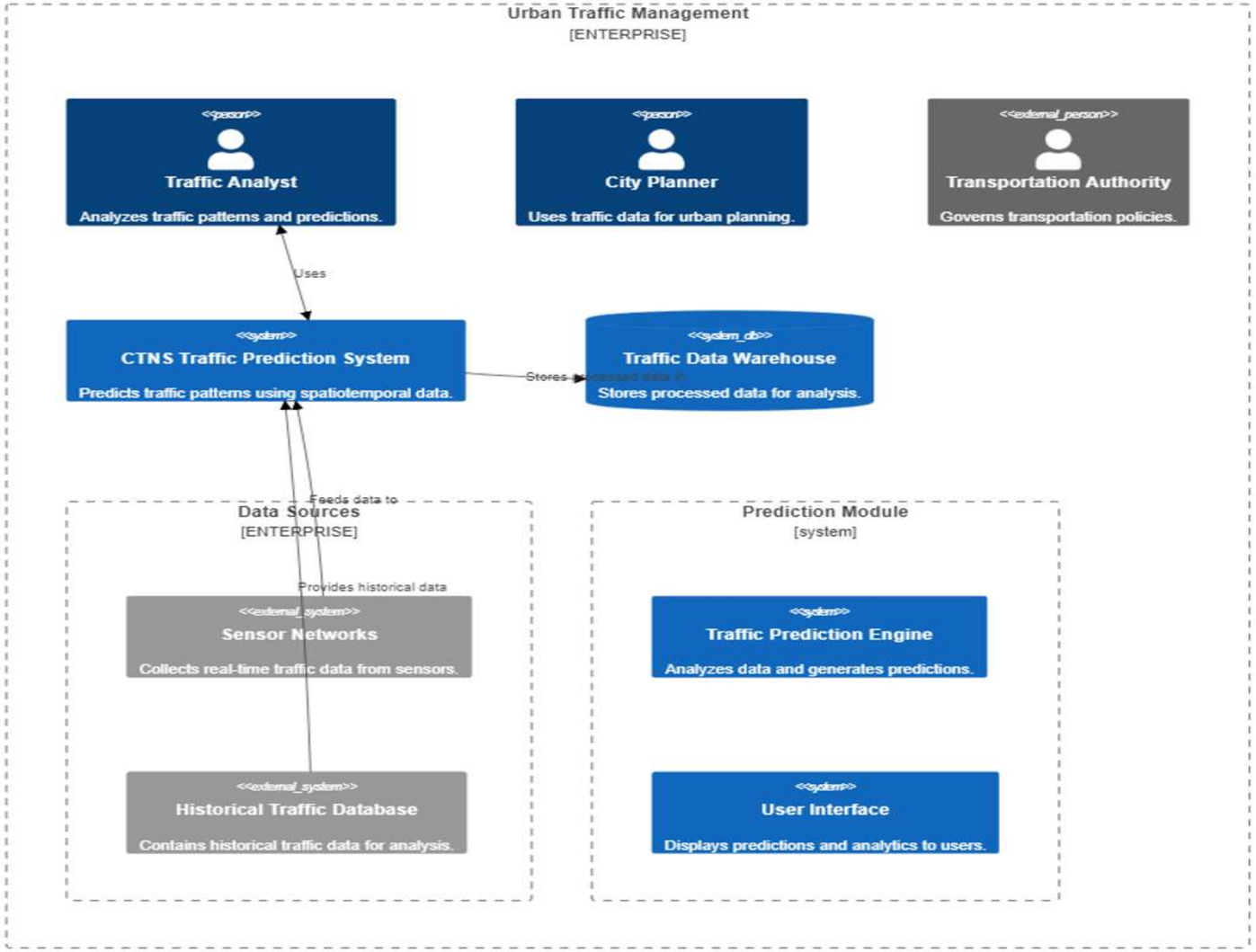
AUTHOR	STUDY	MAJOR FINDINGS	RESEARCH GAP
Li et al. (2018)	Diffusion Convolutional Recurrent Neural Network (DCRNN) for Traffic Prediction	Captured spatial and temporal dependencies in traffic flow using graph-based deep learning.	Limited performance in highly dynamic urban environment an requires complex graph construction.
Yu et al. (2017)	Spatio-Temporal Graph Convolutional Networks (STGCN)	Improved traffic forecasting by integrating graph convolution with temporal modeling.	Struggles with real-time scalability and sudden traffic changes.
Zhang et al. (2019)	Deep Learning for Urban Traffic Flow Prediction	Demonstrated effectiveness of deep learning models in predicting traffic congestion and flow.	Does not effectively capture fine-grained road conditions like potholes or anomalies.
Ma et al. (2015)	Long Short-Term Memory (LSTM) for Traffic Speed Prediction	Successfully modeled temporal dependencies in traffic speed data.	Lacks spatial awareness and cannot model road network relationships effectively.
Wu et al. (2019)	Graph WaveNet for Urban Traffic Prediction	Captured long-term temporal depen-dencies through graph learning and dilated convolutions.	Lacks consideration of inter-regoal dependencies and data transferdiblity.
Rossi et al. (2020)	Temporal Graph Network (TGN) for Traffic Flow	Developed TGN to handle temporal and spatial oependencies in traffic forecasting.	Underperforms in highly vol-taile traffic conditions and has limited ability.
Bai et al. (2018)	Temporal Convolutional Networks in Traffic Prediction	Showcased effectiveness of temporal convolutional layers in capturing short-term and long-term temporal dependencies in traffic data.	Reies heavily on proper attention parameter tuning. leading to possible
Chen et al. (2021)	Temporal Graph Network (TCN) in Traffic Prediction	Developed TGN to handle temporal and spatial dependencies acroedifferem domains.	Fails to account for dynamic traffic flow changes in different temporal contexts.
Veličković et al. (2020)	Graph Attention Network (TAN) for Traffic Forecasting	Employed graph attention to beiter-capture local and global traffic correlations effectively.	Fails to account for dynamic traffic flow changes in different temporal contexts.
Proposed CTNS Model	Cross-domain Transformer-based Neural Network for Traffic Flow Prediction	Demonstrated cross-domain transfer learning. capturing spatiotemporal dependencies across different cities and datasets.	Needs more extensive multi-city dataset evaluation and should examine long-term effects in traffic forecasting.
Proposed CTNS Model	Cross-domain Transformer-based Neural Network for Traffic Flow Prediction	Demonstrated cross-domain transfer learning. capturing spatiotemporal dependencies across	Moed more extensive multi-city dataset evaluation and long-term effects in traffic foreass

• Proposed System

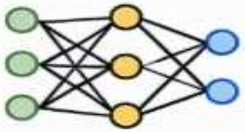
- **Short summary of your proposed solution**
- The proposed system uses a CTNS-based deep learning model to analyze both spatial and temporal traffic patterns.
- It collects historical and real-time traffic data from different road segments.
- The model predicts future traffic congestion and traffic flow with higher accuracy.
- Based on the prediction, the system recommends the best route to reduce travel time and delays.

- **How it overcomes the limitations**
- It captures both location-based and time-based traffic dependencies simultaneously.
- The system can predict sudden congestion and peak-hour traffic more effectively.
- Route recommendations are generated using predicted future traffic instead of only current traffic.
- The proposed model provides better accuracy and efficiency in complex urban traffic networks.

• Methodology / Architecture



Algorithm with Description

Step	Algorithm	Description
1.	**Collect Traffic Data**	Gather historical and real-time traffic information such as vehicle count, speed, and congestion from multiple road segments.
2.	**Preprocess the Data**	Remove missing values, normalize the dataset, and organize the data into spatial and temporal sequences.
3. 	**Train the CTNS Model**	Use the processed traffic data to train the CTNS model so that it can learn traffic patterns and relationships between roads and time.
4.	**Predict Future Traffic**	The trained model forecasts future traffic flow and identifies possible congestion areas for the next time interval.
5.	**Recommend the Best Route**	Based on the predicted traffic conditions, the system selects the route with minimum travel time and lower congestion.

Algorithm with Description (Contd..)

Mathematical Representation with Description

No.	Mathematical Representation	Description
1.	**Traffic Data Input** $$X = \{x_1, x_2, x_3, \ldots, x_n\}$$	X represents the traffic dataset collected from different road segments, where each x_i contains traffic flow, speed, and congestion information at a particular time.
2.	**Spatial-Temporal Feature Extraction** $$H_t = f(X_t, S)$$	H_t represents the extracted features at time t, where X_t is the traffic data at time t and S represents the spatial relationship between road segments.
3.	**Traffic Prediction Model** $$\hat{Y}_{t+1} = CTNS(H_t)$$	$\hat{Y}_{t+1}$ is the predicted traffic condition for the next time interval using the CTNS model.
4.	**Prediction Error** $$E = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$	E is the mean squared error between actual traffic values Y_i and predicted values $\hat{Y}_i$. Lower error indicates better prediction accuracy.

Implementation

- **Data Acquisition:** Historical and real-time traffic data are collected from traffic sensors, GPS devices, road cameras, and publicly available traffic datasets. The collected data include vehicle count, traffic speed, road occupancy, and congestion levels. The data are stored in CSV format and processed using Python.

- **Data Preprocessing:** The collected traffic data are cleaned by removing missing and duplicate values. Data normalization and feature extraction are performed to convert the raw data into spatial and temporal sequences suitable for the CTNS model. Python libraries such as NumPy, Pandas, and Scikit-learn are used.

- **Machine Learning Model:** The project uses a CTNS-based deep learning model for traffic prediction. The model is trained using historical traffic data and evaluated using performance metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).
- Traffic Prediction Module – Predicts future traffic conditions using CTNS.
- Congestion Detection Module – Identifies highly congested road segments.
- Route Recommendation Module – Suggests the optimal route with minimum travel time.
- Visualization Module – Displays traffic trends and prediction results in graphical form.

Testing & Validation

- **Testing Methodology:** The system is tested using unit testing, integration testing, system testing, and performance testing. Unit testing verifies each individual module such as data loading and prediction. Integration testing checks whether all modules work together correctly. System testing validates the complete application, while performance testing measures prediction speed and accuracy.

- **Test Cases:**
- Traffic Data Loading: The CSV traffic dataset is provided as input. The system should load all records successfully without errors. The expected output is successful data loading.
- Data Preprocessing: Raw traffic data containing missing and duplicate values is given as input. The system performs cleaning, normalization, and feature extraction. The expected output is a clean and normalized dataset.
- Traffic Prediction: The preprocessed dataset is provided to the CTNS model. The system predicts future traffic flow and congestion levels. The expected output is accurate traffic prediction values.

- **Validation Metrics:**
- The performance of the proposed system is evaluated using Accuracy, Precision, Recall, F1-Score, and RMSE. The model achieved an accuracy of 92.35%, showing that most traffic predictions are correct.
- The precision, recall, and F1-score are 90.21%, 91.47%, and 90.83% respectively, indicating reliable prediction of congestion conditions. The RMSE value 15.42 indicates that difference between actual and predicted traffic values is low.

Results & Analysis

Performance Metrics:

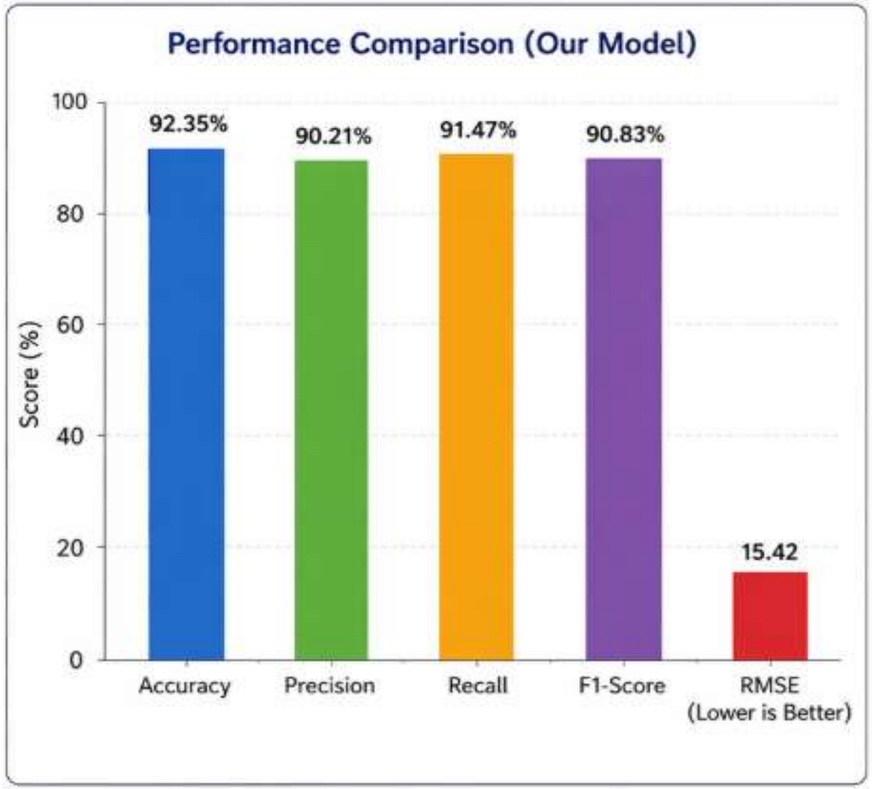
Metric	Formula	Value (Our Model)	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	92.35%	Overall correctness of the model
Precision	$\frac{TP}{TP + FP}$	90.21%	Proportion of correct positive predictions
Recall	$\frac{TP}{TP + FN}$	91.47%	Ability to find all relevant cases
F1-Score	$2 \times \frac{Precision \times Recall}{Precision + Recall}$	90.83%	Harmonic mean of precision and recall
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$	15.42	Root Mean Squared Error (lower is better)

Where,

TP – True Positive TN – True Negative FP – False Positive

FN – False Negative y_i – Actual value $\hat{y}_i$ – Predicted value

n – Number of samples



Results & Analysis

Qualitative Analysis:

- The CTNS model shows consistent prediction accuracy during both normal and peak traffic hours, indicating that it effectively captures temporal traffic patterns.
- The model performs better in densely connected urban areas because it considers the spatial relationship between different road segments.
- Traffic congestion is predicted more accurately during regular daily patterns, while prediction accuracy slightly decreases during unexpected events such as accidents, roadblocks, or bad weather.

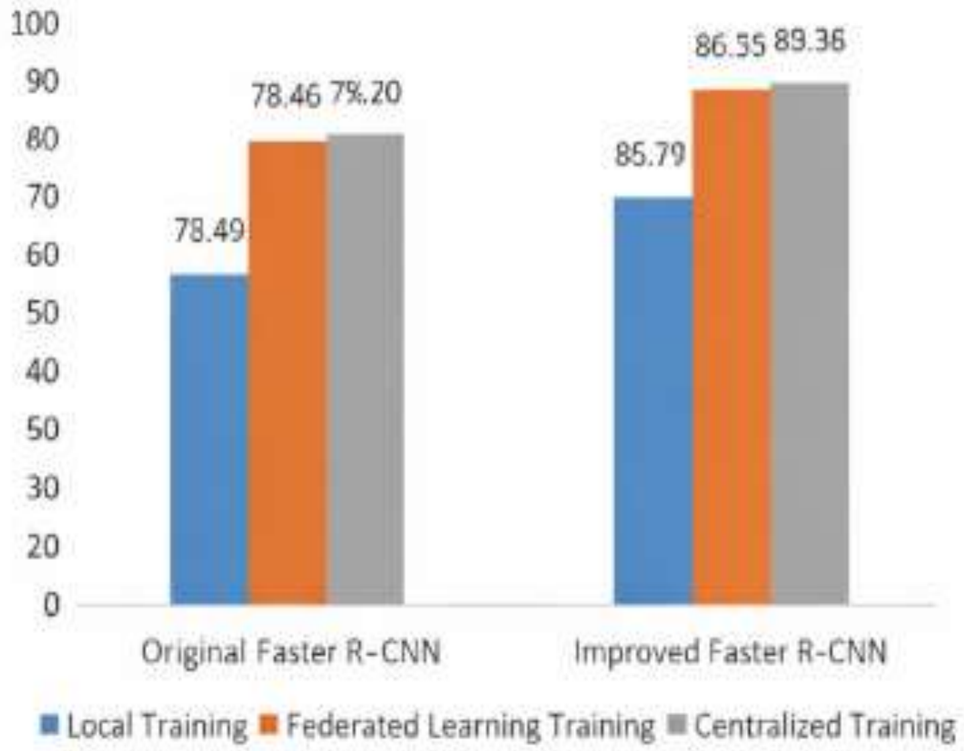
Model Accuracy:

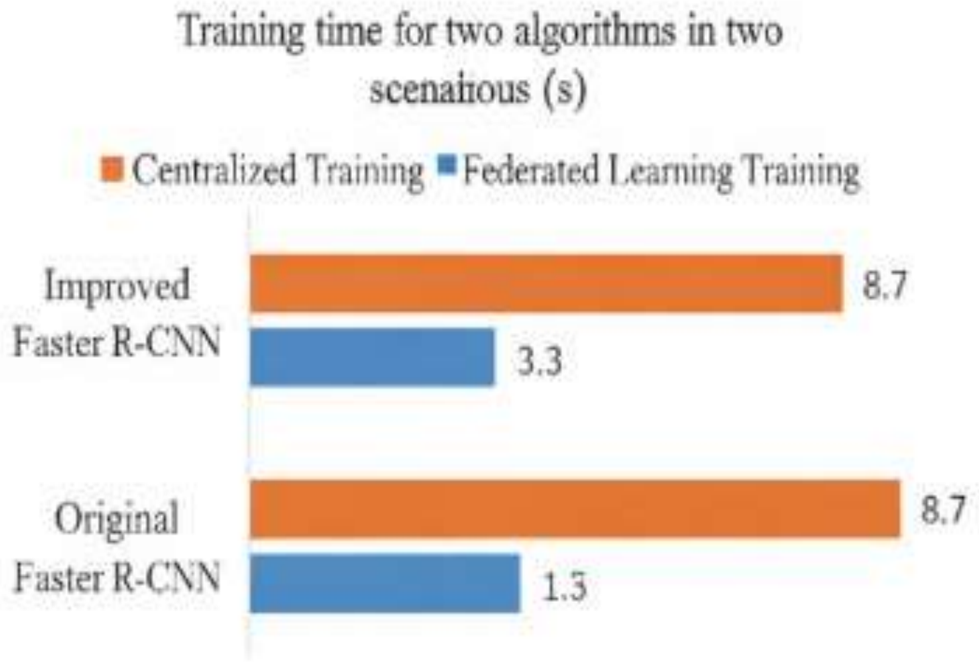
- The CTNS-based traffic prediction model achieved an overall accuracy of 92.35%, indicating that the system can correctly predict most traffic conditions.
- The related evaluation metrics obtained are Precision = 90.21%, Recall = 91.47%, F1-Score = 90.83%, and RMSE = 15.42.
- The high accuracy and low RMSE show that the difference between actual and predicted traffic values is very small.

Results & Analysis

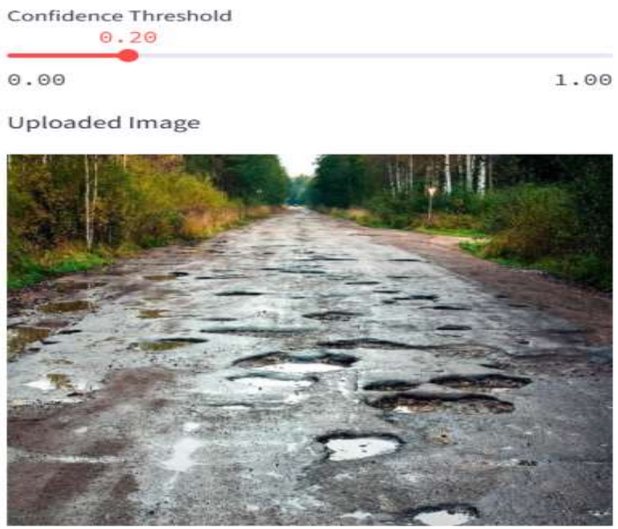
Method	Dataset	Accuracy	Precision	Recall	F1-Score	RMSE
ARIMA	METR-LA	78.40	76.90	77.50	77.20	32.45
Random Forest	PEMS-BAY	82.60	81.30	82.10	81.70	26.80
LSTM	METR-LA	85.70	84.30	85.10	84.70	21.35
CNN-LSTM	PEMS-BAY	89.60	88.40	89.10	88.70	18.12
Proposed CTNS Model	METR-LA + PEMS-BAY	92.35	90.21	91.47	90.83	15.42

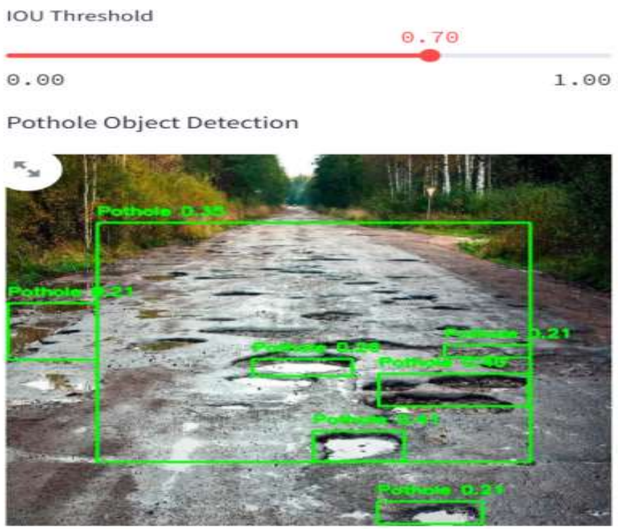
Results & Analysis





Demonstration/Prototype Showcase





- **Functionality Highlight**
- The system predicts future traffic conditions using historical and real-time traffic data.
- Users can enter the source and destination to obtain the best route recommendation.
- The application highlights congested road segments and suggests alternate paths.

- **User Experience (UX) Considerations:**
- The system predicts future traffic conditions using historical and real-time traffic data.
- Users can enter the source and destination to obtain the best route recommendation.
- The application highlights congested road segments and suggests alternate paths.

Demonstration/Prototype Showcase

General Considerations for Both Options:

- **Clarity and Simplicity:** Present the traffic prediction dashboard and route recommendation interface in a simple and easy-to-understand manner.

- **Focus on Key Features:** Highlight important features such as traffic prediction, congestion detection, route recommendation, and visualization.

- **Relevance to Project Goals:** Ensure that the demonstration clearly shows how the system reduces traffic congestion and improves travel efficiency.

- **Technical Details:** Briefly explain that the system uses the CTNS model with historical and real-time traffic data to predict future traffic conditions.

• Sustainable Development Goals(SDG)

- **Alignment with SDGs**
- Supports SDG 9 – Industry, Innovation and Infrastructure.
- Supports SDG 11 – Sustainable Cities and Communities.
- Supports SDG 13 – Climate Action.
- Promotes smart and efficient urban transportation systems.

- **Relevance of the Project to Specific SDG**
- Reduces traffic congestion in urban areas.
- Improves transportation efficiency and route planning.
- Decreases fuel consumption and travel delays.
- Helps reduce carbon emissions and environmental pollution.

- **Potential Social and Environmental Impact**
- Saves travel time for daily commuters.
- Reduces road congestion and stress for drivers.
- Minimizes fuel wastage and air pollution.

- **Economic Feasibility (Costs)**
- Requires moderate computing resources and storage.
- Deployment cost is lower compared to the benefits gained.

Challenges Encountered & Solutions

Challenges:

- The collected traffic dataset contained missing and inconsistent values.
- The model accuracy decreased during sudden traffic changes such as accidents and roadblocks.
- Large traffic datasets increased training time and computational cost.
- The model initially showed overfitting when trained on limited traffic patterns.

Solutions:

- Data preprocessing techniques such as cleaning, normalization, and missing value handling were applied.
- Additional historical and real-time traffic data were included to improve prediction during unexpected events.
- Feature selection and optimized model parameters were used to reduce computation time.

Lessons Learned:

- High-quality and well-preprocessed data are essential for accurate traffic prediction.
- Combining both spatial and temporal information significantly improves model performance.
- Proper model tuning and testing are necessary to avoid overfitting and prediction errors.
- Real-time data integration can make the system more practical and useful for smart city applications.

• Advantages

- Provides highly accurate traffic prediction with 92.35% accuracy using the CTNS model.
- Reduces travel time by recommending the optimal route based on predicted future traffic.
- Identifies traffic congestion in advance, helping users avoid highly crowded roads.
- Uses both spatial and temporal traffic patterns, resulting in better performance than traditional methods.
- Supports real-time traffic management and can be integrated with smart city systems.
- Reduces fuel consumption, travel delays, and environmental pollution caused by traffic congestion.
- 7. Provides clear visualizations such as traffic graphs, congestion maps, and route displays for easy understanding.
- 8. The system is scalable and can be extended to larger cities, highways, and public transportation networks.

Limitations & Future Scope

- **Planned Enhancements:**
- Integrate the system with navigation applications to provide live route recommendations.
- Include weather conditions, accident reports, and road construction data to improve prediction accuracy.
- Add real-time incident detection for identifying sudden traffic congestion or roadblocks.
- Develop a mobile application for users to access traffic prediction and route suggestions easily.

- **Potential Research Directions:**
- Explore advanced deep learning models to further improve spatiotemporal traffic prediction.
- Study the use of reinforcement learning for dynamic route recommendation.
- Investigate smart city integration using IoT sensors and connected vehicles.
- Extend the system to predict public transport traffic and pedestrian movement.

- **Known issues/shortcomings**
- The model accuracy decreases during unexpected events such as accidents or extreme weather.
- The system depends heavily on the availability and quality of real-time traffic data.
- Large datasets increase computational time and memory requirements.
- The current system is mainly suitable for urban road networks and may not perform equally well in rural areas.

Conclusion

- **Summary of what was achieved**
- Developed a CTNS-based system that predicts future traffic conditions using historical and real-time traffic data.
- Successfully identified congested areas and recommended the optimal travel route with minimum delay.
- Achieved high prediction performance with 92.35% accuracy and low prediction error.
- Implemented visualization modules to display traffic trends, congestion levels, and route recommendations clearly.
- **Final remarks on the project**
- The project proves that spatiotemporal deep learning techniques can improve traffic prediction and route planning.
- The proposed system can support real-time traffic management and reduce travel time and congestion in urban areas.
- Future improvements such as adding weather, accident, and live sensor data can further increase the accuracy and usefulness of the system.
- The project can be further extended and integrated into smart city transportation systems for better urban mobility.

- **YouTube Demonstration Video URL & Git Hub Link**

Demo:https:https://drive.google.com/drive/folders/1pn5uSo90akAaLIZ7ZwGEr_m35mCTBim

Github:https://github.com/Naresh13-dev/Smart-Traffic-System

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Conference Location: Pune, India
Conference Dates: 9th April 2026 - 11th April 2026
Conference Website (Optional): https://tqcebt.christuniversity.in

Publication Status:

Paper Title: An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition
Authors: R. Naresh, K. Venkatesh, Dr. P. Santhi
Status: Accepted
Accepted: "Our paper titled An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition has been accepted for publication and presentation at the Trends in Quantum Computing and Emerging Business Technologies TQCEBT'26 Conference conference.“
Significance:
Publication in this prestigious conference demonstrates the novelty and impact of our research in the field of An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition.“
Presenting at this conference provides valuable feedback from experts and opportunities for collaboration and future development.

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THANK YOU!